

CoMa: Contextual Massing Generation with Vision-Language Models

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Figure 1. A complete sample from the CoMa-20K dataset, illustrating the massing generation task. The input defines the problem through multiple modalities: the functional and economical requirements specify the building program, the site contour defines the legal boundaries for development, and the contextual views describe the urban environment. The ground truth output is the 3D massing geometry that satisfies these inputs. In this visualization, the generated massing (in gray) is shown integrated into its urban context (in gray-blue), demonstrating a plausible and context-aware design solution.

Abstract

The conceptual design phase in architecture and urban planning, particularly building massing, is complex and heavily reliant on designer intuition and manual effort. To address this, we propose an automated framework for generating building massing based on functional requirements and site context. A primary obstacle to such data-driven methods has been the lack of suitable datasets. Consequently, we introduce the CoMa-20K dataset, a comprehensive collection that includes detailed massing geometries, associated economical and programmatic data, and visual representations of the development site within its existing urban context. We benchmark this dataset by formulating massing generation as a conditional task for Vision-

Language Models (VLMs), evaluating both fine-tuned and large zero-shot models. Our experiments reveal the inherent complexity of the task while demonstrating the potential of VLMs to produce context-sensitive massing options. The dataset and analysis establish a foundational benchmark and highlight significant opportunities for future research in data-driven architectural design.

1. Introduction

The initial stage of architectural design, known as massing, involves defining the overall three-dimensional shape and volume of a building. This phase is critical as it establishes the project’s relationship with its site, its functional orga-

nization, and its fundamental economic viability. Traditionally, generating and evaluating massing options is a manual, time-consuming process that relies heavily on the designer’s intuition and experience. This reliance creates a bottleneck in the design workflow and can limit the exploration of a wide range of contextually sensitive and programmatically optimized solutions.

While computational methods have evolved from algorithmic optimization to data-driven and generative AI approaches, existing techniques cannot simultaneously handle functional requirements, geometric precision, and visual context. The absence of suitable multi-modal datasets has hindered progress toward unified massing generation systems.

This paper presents three key contributions: (1) **CoMa-20K**, a novel dataset with 20,000 sites combining massing geometries, functional requirements, and urban context views; (2) a **VLM-based benchmark** formulating massing as a conditional generation task, evaluated through both fine-tuning and zero-shot approaches; and (3) a **comprehensive evaluation** revealing fundamental challenges in geometric reasoning and contextual adaptation.

Our work establishes a foundation for data-driven architectural design, demonstrating both the feasibility and complexity of automated massing generation.

2. Related works

Algorithmic and Evolutionary Methods. A significant body of work focuses on generating massing through parametric modeling and evolutionary algorithms. [5, 18, 20] exemplify this approach, using predefined geometric rules and performance criteria (e.g., solar exposure, energy efficiency) to generate and optimize massing forms. While these methods are powerful for navigating a defined solution space towards specific objectives, they lack the flexibility to learn implicit, real-world design patterns from data. Their output is constrained by the initial rule set and requires significant expert knowledge to set up, making them less adaptable to novel contextual or programmatic cues compared to data-driven methods.

Agentic and Collaborative AI Approaches. With the rise of large language models, recent work explores using AI as a collaborative agent in the design process. [3, 22] demonstrate frameworks where multimodal AIs converse with human designers to support early-stage exploration. These systems excel at ideation and leveraging common-sense knowledge but operate in a text-centric manner without being grounded in a large-scale, structured dataset of architectural forms and their real-world constraints. Consequently, they lack the capability for direct, end-to-end massing generation from specified site conditions.

Data-Driven Deep Learning Methods. Several studies have applied deep learning directly to geometric gen-

eration. [15] uses a Deep Neural Network to classify and investigate high-rise building types, but does not generate geometry. [19] proposes I2bnet, which generates a BIM massing model from a single image, demonstrating a data-driven approach. Similarly, [16] integrates footprint prediction with massing in a Pittsburgh case study. While these works share our data-driven philosophy, they typically do not condition generation on rich, multi-modal inputs (textual requirements and visual context) simultaneously, don’t use Vision-Language Models [2], and they operate on a scale and modality different from our VLM-based formulation.

Contextual and Urban-Scale Generation. Research in this area focuses on generating building geometries that respond to their environment. [16] integrates building footprint prediction with massing in a Pittsburgh case study, creating context-aware massing models, but relies on traditional GIS data and parametric modeling rather than learning from a large-scale dataset. [22] presents a framework combining computational optimization with generative AI for performative design, yet it operates at a conceptual level using text-to-image models and does not produce structured, parseable 3D massing geometry. In contrast, our work introduces a dedicated dataset and an end-to-end VLM framework that directly generates detailed, program-specific 3D massing from structured requirements and visual context, moving beyond conceptual inspiration or parametric constraints to data-driven synthesis.

3. Dataset

We introduce the CoMa-20K dataset, a novel and large-scale collection designed specifically for the task of contextual building massing generation. The dataset associates massing geometry within a defined site contour with corresponding functional-economical requirements and multi-view imagery of the surrounding urban context. The following subsections detail the data collection methodology, structural properties, and preprocessing steps undertaken to construct this resource.

3.1. Dataset Overview

The CoMa-20K dataset is a multi-modal collection where each sample represents a unique development site and provides all necessary data to formulate the massing generation task. Each sample contains the associated massing geometry, functional-economical requirements, and visual context (see Figure 1). The dataset is designed to learn the complex relationships between a site’s constraints, its urban context, and the resulting building forms.

A single data sample in the CoMa-20K dataset contains the following core components:

- **site_contour:** A closed polygon defining the boundaries of the development site, provided as a list of $[x, y]$

Table 1. Description of the attributes within the `requirements` dictionary.

Attribute	Data Type	Description
<code>id</code>	String	Unique identifier for the building, corresponding to an entry in the <code>massing</code> list.
<code>building_function</code>	String	The primary function of the building (e.g., “Commercial Accommodation”).
<code>n_dwelling</code> s	Float	The total number of living spaces (apartments, rooms, etc.).
<code>building_name</code>	String	The name of the building, if one exists.
<code>n_floors</code>	Float	The number of floors above ground.
<code>n_commercial_spaces</code>	Integer	The total count of commercial spaces (e.g., offices, retail) within the building.
<code>offices</code>	List[Dict]	A list of specific commercial spaces. Each item contains a <code>name</code> and a <code>function</code> describing the business.
<code>public_spaces</code>	List[Dict]	A list of spaces with public capacity (e.g., cafes, restaurants). Each item contains <code>name</code> , <code>function</code> , <code>indoor_capacity</code> , and <code>outdoor_capacity</code> .
<code>floor_height</code>	Float	The mean height of a single floor, in meters.
<code>usable_area</code>	Float	The total usable area of the building, calculated as the sum of the areas of all floors, in square meters.

coordinates in meters within a local Cartesian coordinate system.

- **massing:** A list of building massings located within the `site_contour`. Each massing is defined by its unique `id` and a detailed geometric representation.
- **requirements:** A list of functional and economical specifications for each building in the `massing` list. Each requirement dictionary shares the same `id` as its corresponding massing, ensuring a one-to-one mapping.
- **Contextual Images:** A set of five images providing different visual representations of the site and its context:
 - `env_render_high`: A high-detail, photorealistic rendering of the massing within its urban environment.
 - `env_render_low`: A low-detail, schematic rendering of the same context.
 - `env_image`: A raw mesh visualization of the context.
 - `map_image_base` & `map_image_alt`: Top-down map views of the site in two different cartographic styles.

3.1.1. Data Schema and Properties

Massing Geometry: The `massing` property provides a precise geometric definition. Each building massing is composed of a list of horizontal extrusions. Each extrusion is defined by a polygonal footprint (`polygons`), a `bottom_elevation`, and a `top_elevation` (in meters), allowing for the representation of complex, multi-level structures.

Functional-Economical Requirements: The `requirements` property provides a rich set of attributes that define the programmatic and economic intent

behind each building. The structure and description of these attributes are detailed in Table 1.

3.2. Data Sources

The CoMa-20K dataset is constructed entirely from open data provided by the City of Melbourne via their public data platform [4]. This rich source provides a cohesive, real-world foundation for urban modeling. From the approximately 200 available datasets, we selectively utilized seven key tables that, when combined, provide the geometric, contextual, and programmatic information required for our task. The integration of these tables allows us to move from raw civic data to a structured dataset for generative AI.

The data synthesis process revolves around the development site, defined by property boundaries. The following tables were used, with their roles detailed below:

- **Property Boundaries:** This table serves as the foundational layer for the entire dataset. It provides the unique site identifier (`id`) and the precise polygonal `site_contour` for each development lot. All other data is aggregated and associated with these defined sites.
- **2023 Building Footprints:** This table supplies the raw geometric data for constructing the 3D massing models and the surrounding urban context. It contains individual footprint polygons, each with an extrusion height. As these footprints lack a direct property identifier, they are spatially joined to the `site_contour` to assemble complete building massings and to the CLUE blocks to build the environmental context.
- **Blocks for Census of Land Use and Employment (CLUE):** These polygons define small statistical districts

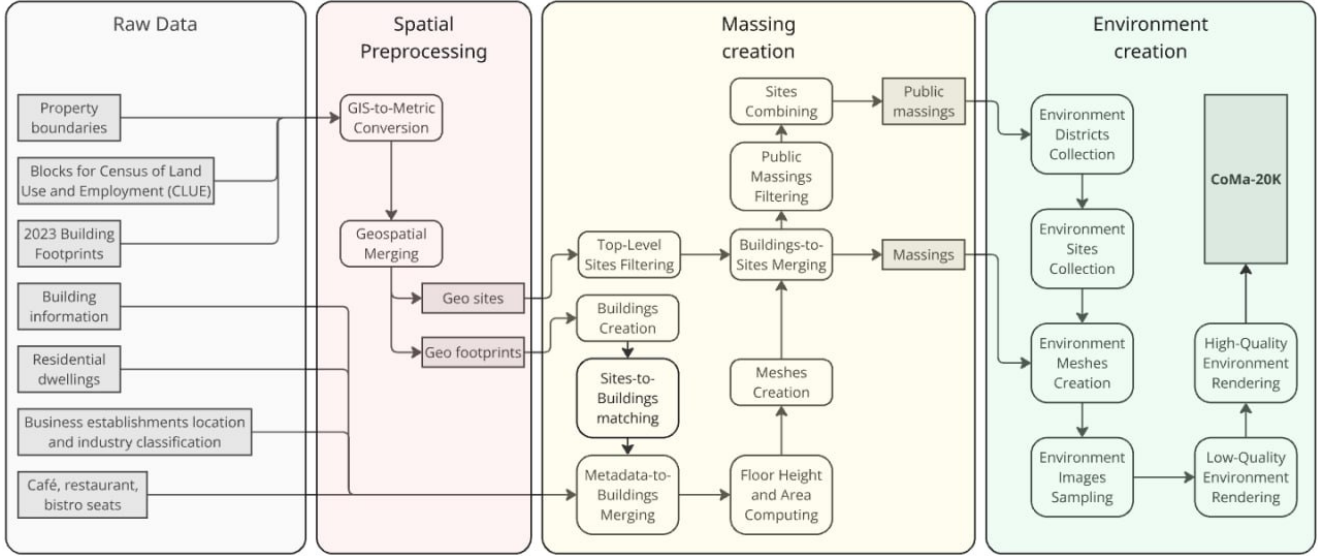


Figure 2. Dataset Creation Pipeline. The pipeline for constructing the CoMa-20K dataset consists of three main stages: Spatial Preprocessing, where raw geospatial data is converted to a metric system and prepared for merging; Massing Creation, where building geometries are assembled and fused with their functional-economical metadata; and Environment Creation, where the visual context for each site is generated by assembling neighboring buildings and producing multi-style renderings.

within the city. We use them as a practical spatial unit to sample the immediate surrounding environment of a target site, ensuring the contextual views are representative of the local urban fabric.

The following tables are linked directly to a development site via a shared property ID, enabling the compilation of the functional and economical requirements:

- **Building Information:** This is a primary source for key architectural attributes, providing data such as `building_name`, `n_floors`, `building_function`, and `floor_height` for structures on a given property.
- **Residential Dwellings:** For buildings with a residential function, this table provides the crucial `n_dwellings` attribute, detailing the number of living spaces (e.g., apartments, houses) within a building on a specific property.
- **Business Establishments Location and Industry Classification:** This table is used to populate the `offices` list within the requirements. It provides a record of commercial tenants within a building, including their name and business function.
- **Café, Restaurant, Bistro Seats:** Despite its name, this table catalogs a wide range of public spaces with known public capacity. It provides the name, function, `indoor_capacity`, and `outdoor_capacity` for venues like cafes, restaurants, and bars located on a property.

3.3. Dataset Collection Pipeline

The construction of the CoMa-20K dataset from the raw open data sources is a multi-stage process. This pipeline transforms disparate geospatial and tabular data into a cohesive, multi-modal dataset suitable for training vision-language models. The entire procedure can be divided into three sequential stages: Spatial Preprocessing, Massing Creation, and Environment Creation.

Spatial Preprocessing

This initial stage prepares the raw geospatial data for subsequent geometric operations and merging.

- **GIS-to-Metric Conversion:** All polygon-based datasets (Property Boundaries, Building Footprints, CLUE Blocks), which are originally provided in longitude-latitude coordinates (EPSG:4326), are projected into a local metric coordinate system. This conversion is critical for performing accurate area calculations and spatial intersections.
- **Geospatial Merging:** To link datasets that lack a common identifier (e.g., linking Footprints to Property Boundaries), an all-to-all spatial intersection is computed. To ensure robustness against numerical errors, we do not use a simple binary intersection. Instead, we calculate the actual area of intersection, using this as a reliable metric for establishing relationships between polygons.

Massing Creation

This core stage focuses on constructing the 3D building massings and associating them with their functional-

economical requirements.

- **Top-Level Sites Filtering:** The Property Boundaries dataset contains nested sites (e.g., a private lot within a larger development). We filter to retain only the top-level, physically distinct development sites to avoid duplication and ensure each site represents a coherent area for development.
- **Buildings Creation:** Individual footprint extrusions from the Building Footprints table are merged together based on spatial proximity to form complete building massings.
- **Sites-to-Building Matching:** Each constructed building is matched to its corresponding development site. A building is assigned to a site if the area of intersection between the building’s footprint and the site polygon constitutes over 90% of the building’s total footprint area, ensuring a strong and unambiguous spatial relationship.
- **Metadata-to-Buildings Merging:** Using the established site-to-building links, we merge all metadata from the Building Information, Residential Dwellings, Business Establishments, and Public Spaces tables via their shared property ID. This aggregated data forms the comprehensive requirements dictionary for each building.
- **Floor Height and Area Computing:** Key physical parameters are calculated directly from the geometry. The mean `floor_height` is derived from the extrusion data and the number of floors. The `usable_area` is computed as the sum of the areas of all horizontal extrusions (floors) within the building massing.
- **Meshes Creation:** The geometric data for each building is compiled into a standard 3D mesh format for rendering and visualization.
- **Buildings-to-Sites Merging:** The finalized buildings, now with their complete geometry and requirements, are grouped back together by their associated site to form the core `massing` and `requirements` lists for each dataset sample.
- **Public Massings Filtering:** To focus on larger-scale development, we filter out private houses and townhouses, retaining only public-facing properties such as apartment complexes and commercial offices.
- **Sites Combining:** To increase dataset diversity and include sites with multiple buildings, we programmatically combine 2 to 4 neighboring properties into larger, aggregated development sites.

Environment Creation

This final stage generates the crucial contextual visual data for each development site.

- **Environment Districts Collection:** For each target site, we identify its parent CLUE block and all immediately adjacent (neighboring) blocks. Adjacency is determined by a boundary intersection, ensuring a zero-area touch.
- **Environment Sites Collection:** All development sites lo-

cated within these identified environment districts (parent and neighbors) are collected to form the “context” for the target site.

- **Environment Meshes Creation:** A unified 3D environment mesh is created for each target site by concatenating the meshes of all buildings within the collected environment sites.
- **Environment Images Sampling:** Single view for each environment mesh is generated. The target site’s contour is overlaid in red for visibility. The camera pose is optimized to maximize the number of pixels occupied by the target site, ensuring it is prominently visible in the context. In parallel, top-down `map_image` views are sampled with the site polygon highlighted in two different cartographic styles.
- **Low-Quality Environment Rendering:** The raw environment mesh view is passed through the *Qwen-Image-Edit* [14] model to semantically enhance the scene by adding realistic elements like roads and parks, producing a low-realism city environment image (`env_render_low`).
- **High-Quality Environment Rendering:** The low-quality render is fed again into the same image editing model with a prompt for photorealism, resulting in the final, high-detail contextual visualization (`env_render_high`).

3.4. Statistics and Splitting

The final CoMa-20K dataset comprises 20,000 unique development sites, each with associated massings, requirements, and contextual imagery. To facilitate robust model training and evaluation, the dataset was randomly split into a training/validation set (90%, 18,000 samples) and a held-out test set (10%, 2,000 samples). This ensures that the model is evaluated on a statistically representative sample of the data that it did not see during training.

An analysis of the dataset reveals several key distributions that characterize the urban fabric of Melbourne as captured in our data:

- **Building Function:** The most prevalent building type is “Residential Apartment,” reflecting the city’s dense urban core.
- **Buildings per Site:** The majority of development sites contain between 1 and 4 distinct buildings, with single-building sites being the most common.
- **Dwelling Units:** While the mode is a single dwelling unit per building (corresponding to many commercial or small-scale structures), there is a significant secondary mode of buildings containing between 10 and 50 units, representative of mid-rise apartment complexes.
- **Commercial Spaces:** The number of offices or commercial spaces within a building is generally low, with most buildings containing fewer than 5. However, the distri-

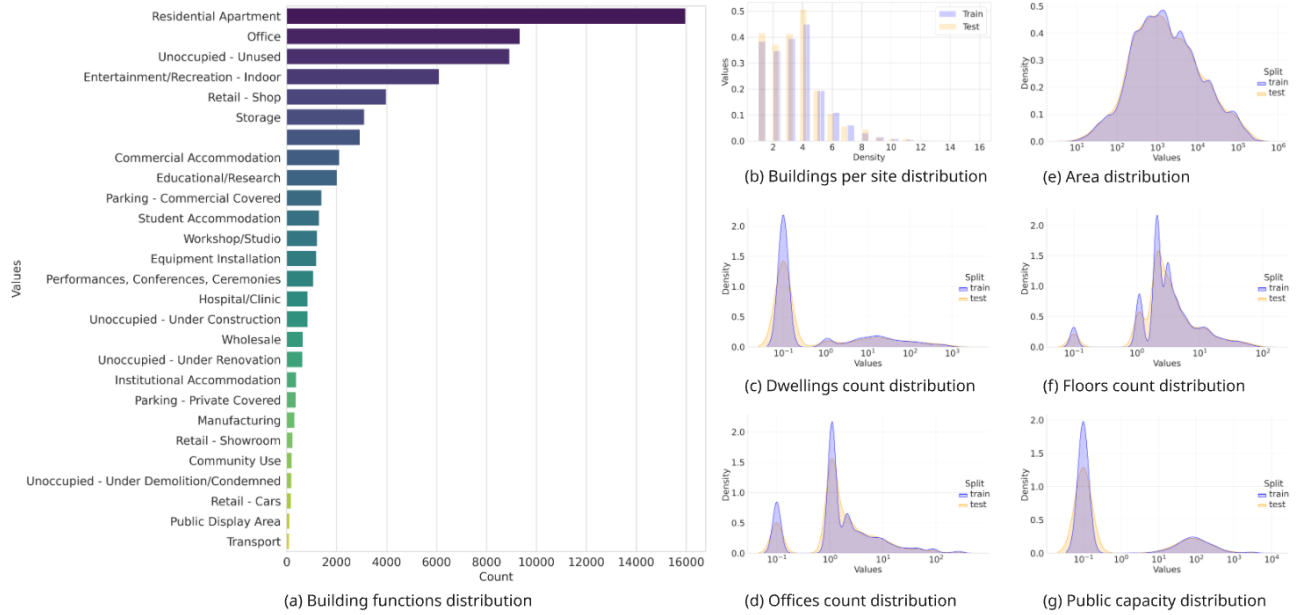


Figure 3. CoMa-20K Dataset Statistics. Distributions of key attributes across the dataset: (a) building functions, (b) number of buildings per site, (c) dwelling units per building, (d) commercial spaces per building, (e) total usable area, (f) number of floors, and (g) public capacity of commercial spaces.

bution has a long tail, extending up to approximately 100 for large commercial towers.

- **Total Building Area:** The distribution of `usable_area` is right-skewed, with a concentration of buildings around 1,000 m² and a long tail of larger structures.
- **Building Height:** The number of floors is heavily concentrated in low-rise buildings (less than 10 floors), with a small but significant number of high-rise buildings present in the dataset.
- **Public Capacity:** For the subset of buildings that have this data, the public capacity (sum of `indoor_capacity` and `outdoor_capacity`) is typically distributed around 100 people per building.

Detailed distributions for these and other attributes are provided in Figure 3, which offers a comprehensive visual summary of the dataset’s composition.

4. Method

We frame the problem of building massing generation as a conditional vision-language task. The objective is to generate a 3D massing model conditioned on a textual description of its functional-economical requirements, development site contour, and its urban visual context.

4.1. Finetuned VLM Approach

We fine-tune the Qwen3-VL series [1] of Vision Language Models on our CoMa-20K dataset. The

model input is a concatenated prompt consisting of the site’s requirements formatted as a text string, the `site_contour` coordinates, and a contextual image (`env_render_high`). The training objective is to autoregressively generate the tokenized 3D massing geometry. We experiment with three model variants (2B, 4B, and 8B parameters) to study the scalability of this approach.

4.2. Zero-shot VLM Approach

To benchmark the inherent capability of large VLMs on this task, we employ *Qwen3-VL-235B-A22B-Instruct* [9] in a zero-shot setting. The model is prompted with detailed architectural rules and spatial logic constraints, including:

- **Spatial Logic:** Strict adherence to site boundaries, coordinate system consistency, and geometric containment
- **Placement Logic:** Footprint area estimation, orientation parallel to site contours, and minimum buffer zones
- **Geometric Articulation:** Avoidance of simple rectangles through L/U-shaped plans, step-backs, and varied roof heights
- **Structural Constraints:** Proper elevation calculations based on floor counts and heights

The prompt explicitly defines the input-output format and emphasizes outputting only structured JSON geometry, enabling direct comparison with our fine-tuned approaches while leveraging the model’s pre-trained capabilities.

Table 2. Quantitative Results. Performance comparison of fine-tuned and zero-shot models across key metrics.

	Pattern Match $\uparrow$	JSON Validity $\uparrow$	ID IoU $\uparrow$	Floor Error $\downarrow$	Area Error $\downarrow$	Site IoU $\uparrow$	Contextual Relevance $\uparrow$
CoMa-2B	0.84	0.63	0.46	0.76	1.30	0.01	0.15
CoMa-4B	0.95	0.72	0.71	0.49	2.86	0.03	0.18
CoMa-8B	0.94	0.79	0.75	0.42	1.90	0.05	0.24
Qwen3-VL-235B-Instruct	1.00	0.99	0.99	0.12	0.79	0.10	0.25

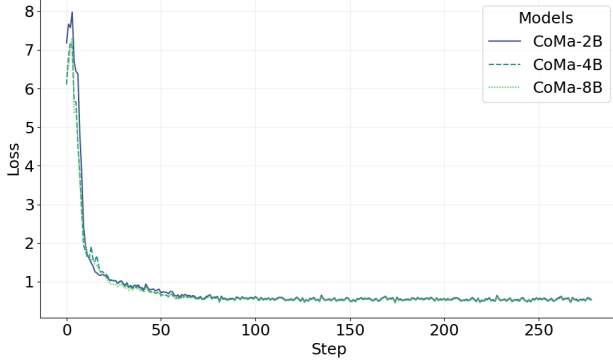


Figure 4. Learning curves for the fine-tuned models.

5. Experiments

5.1. Implementation Details

We conducted fine-tuning experiments on three model variants: *Qwen3-VL-2B-Instruct* [10], *Qwen3-VL-4B-Instruct* [11], and *Qwen3-VL-8B-Instruct* [12]. The training configuration was kept consistent across all models to ensure a fair comparison.

All models were fine-tuned using Low-Rank Adaptation (LoRA) [6], applied to the query, key, and value projection layers within the attention mechanisms [17]. We used the AdamW [8] optimizer with a learning rate of 5×10^{-4} and a cosine learning rate scheduler with a 5% linear warmup phase [7].

Training was performed for 3 epochs on a single node equipped with 8 H100 GPUs. We utilized the Fully Sharded Data Parallel (FSDP) [21] strategy for efficient memory usage and distributed training, with a global batch size of 128.

The training loss curves for all model sizes are provided in Figure 4, demonstrating stable convergence over the course of training.

5.2. Quantitative Results

We evaluate both fine-tuned and zero-shot models using a comprehensive set of metrics designed to assess both the structural correctness and contextual appropriateness of the generated massings. The metrics are defined as follows:

- **Pattern Match:** The rate of model outputs from which a JSON-like string can be successfully extracted.
- **JSON Validity:** The rate of extracted strings that are valid, parseable JSON.
- **ID IoU:** The Intersection-over-Union between the set of building IDs in the requirements and the set of IDs in the generated massing.
- **Floor Error:** The mean absolute difference between the required and predicted number of floors, averaged per site and normalized by the ground truth value. For a site with N buildings, it is computed as:

$$\frac{1}{N} \sum_{i=1}^N \frac{|F_{pred}^i - F_{gt}^i|}{F_{gt}^i} \quad (1)$$

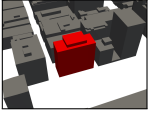
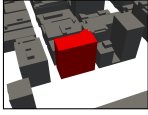
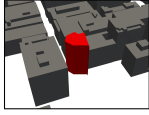
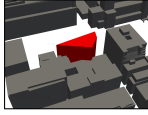
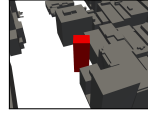
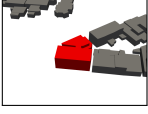
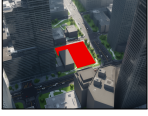
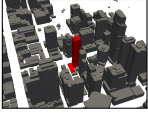
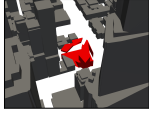
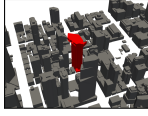
- **Area Error:** The mean absolute difference between the required and predicted usable area, averaged per site and normalized by the ground truth area:

$$\frac{1}{N} \sum_{i=1}^N \frac{|A_{pred}^i - A_{gt}^i|}{A_{gt}^i} \quad (2)$$

- **Site IoU:** The IoU between the ground truth site contour and the union of the bottom polygons of all generated buildings.
- **Contextual Relevance:** A binary score (0/1) assigned by a powerful VLM-as-a-judge (*Qwen3-VL-235B-A22B-Thinking* [13]), prompted with architectural rules to determine if the generated massing is plausible within its surrounding environment.

The quantitative results, presented in Table 2, reveal several key findings. Among the fine-tuned models, the largest 8B parameter model outperforms smaller models across all metrics except Area Error, where the 2B model achieves the best performance. The large zero-shot model demonstrates superior capability compared to small fine-tuned models, achieving near-perfect scores in format-related metrics (Pattern Match and JSON Validity) and showing competitive performance in Floor Error and Area Error. However, despite outperforming all fine-tuned models in Site IoU and Contextual Relevance, the absolute scores for these spatial and contextual metrics remain relatively low, indicating sig-

Table 3. Qualitative Results. Representative examples from fine-tuned and zero-shot models, illustrating differences in geometric complexity, artifact prevalence, and contextual adaptation.

Input			Output (fine-tuned)			Output (zero-shot)
Requirements	Context	Ground Truth	CoMa-2B	CoMa-4B	CoMa-8B	Qwen3-VL-235B
Building function: <i>Educational/Research</i> Floors count: 15 Total area: 15565 Dwellings count: 0 Offices count: 47 Offices public capacity: 68						
Building function: <i>Student Accommodation</i> Floors count: 3 Total area: 1551 Dwellings count: 65 Offices count: 0 Offices public capacity: 0						
Building function: <i>Residential Apartment</i> Floors count: 50 Total area: 36259 Dwellings count: 227 Offices count: 4 Offices public capacity: 0						

nificant challenges in geometric accuracy and contextual integration that persist across all approaches.

5.3. Qualitative Results

Qualitative analysis of the generated massings reveals distinct behavioral patterns between the fine-tuned and zero-shot approaches. As shown in Table 3, the fine-tuned models demonstrate a tendency to generate geometrically complex massings, often featuring articulated forms with multiple extrusions and varied roof heights. However, these complex outputs frequently contain geometric artifacts such as self-intersecting polygons and irregular shapes that deviate from architecturally plausible forms. This phenomenon can be attributed to the dataset’s composition: while containing some highly complex examples, the CoMa-20K dataset is dominated by simpler massings, creating an imbalance that challenges the models’ ability to generalize effectively for complex geometric generation.

In contrast, the large zero-shot model produces geometrically clean and consistent outputs, but exhibits strong limitations in complexity and contextual adaptation. The generated massings are predominantly simple rectangular forms with uniform orientations, demonstrating insufficient consideration of site contour geometry and surrounding building context. This suggests that while the model possesses strong general reasoning capabilities, it lacks specialized geometric understanding and fails to adequately incorporate spatial and contextual cues from the visual input, resulting in massings that, while structurally valid, are architecturally simplistic and contextually disconnected.

6. Conclusion

We introduced CoMa-20K, the first large-scale dataset pairing 3D massing geometries with functional requirements and visual urban context, enabling massing generation as a conditional VLM task. Our benchmark reveals this task’s fundamental challenges: while fine-tuned models attempt complex geometries, they suffer from artifacts, whereas large zero-shot models produce clean but overly simplistic forms lacking contextual adaptation.

These findings establish CoMa-20K as a crucial benchmark and highlight two key research directions: developing specialized architectures for geometric reasoning and creating more balanced datasets. Our work provides a foundation for AI-augmented design tools that bridge programmatic requirements with contextual generation, expanding rather than replacing human creativity.

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